

ENEL700 Tutorial 4

1. An 162MHz carrier is deviated by 12kHz by a 2kHz modulating signal. What is the modulation index? Is the resulting FM signal a narrow band FM signal?

$$\beta = \frac{f_d}{f_m} = \frac{12\text{kHz}}{2\text{kHz}} = 6 > 0.25. \text{ No, it is a WBFM signal.}$$

2. For a single frequency sine wave modulating signal of 3kHz with a carrier frequency of 36MHz, what is the spacing between sidebands? Calculate the bandwidth occupied by the signal using Carson's rule if the frequency deviation is 12kHz.

$$\text{Spacing between sidebands is } f_m = 3\text{kHz}$$

$$\text{Carson's bandwidth} = 2(f_d + f_m) = 2(12\text{kHz} + 3\text{kHz}) = 30\text{kHz}$$

3. For a FM signal with frequency deviation of 8kHz and its modulating signal with frequency of 2kHz, what is the minimum bandwidth of the signal that contains 99% of its signal power?

$$\beta = \frac{f_d}{f_m} = \frac{8\text{kHz}}{2\text{kHz}} = 4 \Rightarrow J_0^2(\beta) + 2J_1^2(\beta) + 2J_2^2(\beta) + \dots + 2J_6^2(\beta) = 0.9944 (>0.99) \text{ when } n=6$$

$$\text{Therefore, the minimum bandwidth: } B = 2nf_m = 2 \times 6 \times 2\text{kHz} = 24\text{kHz}$$

4. A 5kHz sinusoidal signal with amplitude 5V serves as the modulating signal for a FM modulator of sensitivity of 15kHz/V, determine the Carson's bandwidth of the resulting FM signal.

$$\beta = \frac{k_f V_m}{f_m} = \frac{15 \times 5}{5} = 15 \quad \text{Carson's bandwidth} = 2(\beta + 1)f_m = 160\text{kHz}$$

5. A carrier of 49MHz is frequency-modulated by a 1.5kHz square wave. The modulation index is 0.25. Sketch the spectrum of the resulting signal, assuming that only harmonics less than the sixth are passed by the system.

For a 1.5kHz square wave (assuming amplitude 1V), it can be expressed as:

$$v(t) = 1 + \frac{4}{\pi} \left[\cos(2\pi f_0 t) - \frac{1}{3} \cos(2\pi 3f_0 t) + \frac{1}{5} \cos(2\pi 5f_0 t) - \frac{1}{7} \cos(2\pi 7f_0 t) + \dots \right]$$

